Contents

SUGRA Bases with ≥ 6 entries per group

Excluding $\mathfrak{su}_2$, $\mathfrak{su}_3$ externals(which cause bases with large intersection numbers) 77 groups, 505 bases.

$T = 4$, Ext $= \mathfrak{e}_6$, LST $= \mathfrak{su}_3$ (7)

[1] 2131^1

$T = 4,\ \Delta = -243,\ \det = 1,\ (n_+, n_-, n_0) = (1, 4, 0)$

[2] $1\overset{1^1}{3}1$

$T = 4,\ \Delta = -243,\ \det = 1,\ (n_+, n_-, n_0) = (1, 4, 0)$

[3] 131^12

$T = 4,\ \Delta = -243,\ \det = 4,\ (n_+, n_-, n_0) = (1, 4, 0)$

[4] $1\overset{1^1}{3}1^1$

$T = 4,\ \Delta = -243,\ \det = 4,\ (n_+, n_-, n_0) = (1, 4, 0)$

[5] 221^13

$T = 4,\ \Delta = -243,\ \det = 9,\ (n_+, n_-, n_0) = (1, 4, 0)$

[6] 21^131^1

$T = 4,\ \Delta = -243,\ \det = 9,\ (n_+, n_-, n_0) = (1, 4, 0)$

[7] $1^1\overset{1^1}{3}1^1$

$T = 4,\ \Delta = -243,\ \det = 9,\ (n_+, n_-, n_0) = (1, 4, 0)$

$T = 5$, Ext $= \mathfrak{so}_8$, LST $= \mathfrak{so}_8$ (12)

[1] $1\overset{1}{\underset{1^1}{4}}1$

$T = 5,\ \Delta = -184,\ \det = -1,\ (n_+, n_-, n_0) = (1, 5, 0)$

[2] $1\overset{1^1}{4}12$

$T = 5,\ \Delta = -184,\ \det = -1,\ (n_+, n_-, n_0) = (1, 5, 0)$

[3] 22141^1

$T = 5,\ \Delta = -184,\ \det = -1,\ (n_+, n_-, n_0) = (1, 5, 0)$

[4] $1\overset{1^1}{\underset{1^1}{4}}1$

$T = 5,\ \Delta = -184,\ \det = -4,\ (n_+, n_-, n_0) = (1, 5, 0)$

[5] $21\overset{1^1}{4}1^1$

$T = 5,\ \Delta = -184,\ \det = -4,\ (n_+, n_-, n_0) = (1, 5, 0)$

[6] $1\overset{21^1}{4}1$

$T = 5,\ \Delta = -184,\ \det = -4,\ (n_+, n_-, n_0) = (1, 5, 0)$

[7] 21^1412

$T = 5,\ \Delta = -184,\ \det = -4,\ (n_+, n_-, n_0) = (1, 5, 0)$

[8] $1^1\overset{1^1}{\underset{1^1}{4}}1^1$

$T = 5,\ \Delta = -184,\ \det = -16,\ (n_+, n_-, n_0) = (1, 5, 0)$

[9] $21^1\overset{1^1}{4}1^1$

$T = 5,\ \Delta = -184,\ \det = -16,\ (n_+, n_-, n_0) = (1, 5, 0)$

[10] 221^141^1

$T = 5,\ \Delta = -184,\ \det = -16,\ (n_+, n_-, n_0) = (1, 5, 0)$

[11] 21^141^12

$T = 5,\ \Delta = -184,\ \det = -16,\ (n_+, n_-, n_0) = (1, 5, 0)$

[12] $41^{1}222$

$T=5,\ \Delta=-184,\ \det=-16,\ (n_+,n_-,n_0)=(1,5,0)$

$T=7$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{so}_8\oplus\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2$ (6)

[1] 14123211^{1}

$T=7,\ \Delta=-137,\ \det=-1,\ (n_+,n_-,n_0)=(1,7,0)$

[2] 1232141^{1}

$T=7,\ \Delta=-137,\ \det=-1,\ (n_+,n_-,n_0)=(1,7,0)$

[3] $12\overset{1^{1}41}{3}2$

$T=7,\ \Delta=-137,\ \det=-1,\ (n_+,n_-,n_0)=(1,7,0)$

[4] $1\overset{1^{1}}{4}1\overset{2}{3}2$

$T=7,\ \Delta=-137,\ \det=-4,\ (n_+,n_-,n_0)=(1,7,0)$

[5] $1^{1}412321^{1}$

$T=7,\ \Delta=-137,\ \det=-4,\ (n_+,n_-,n_0)=(1,7,0)$

[6] $141\overset{2}{3}21^{1}$

$T=7,\ \Delta=-137,\ \det=-4,\ (n_+,n_-,n_0)=(1,7,0)$

$T=7$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{so}_8^2$ (6)

[1] $1\overset{1^{1}}{4}1\overset{1}{4}1$

$T=7,\ \Delta=-154,\ \det=-1,\ (n_+,n_-,n_0)=(1,7,0)$

[2] $1\overset{1^{1}}{4}1412$

$T=7,\ \Delta=-154,\ \det=-1,\ (n_+,n_-,n_0)=(1,7,0)$

[3] $1\overset{1^{1}}{4}1\overset{1^{1}}{4}1$

$T=7,\ \Delta=-154,\ \det=-4,\ (n_+,n_-,n_0)=(1,7,0)$

[4] $1^{1}\overset{1^{1}}{4}1\overset{1^{1}}{4}1^{1}$

$T=7,\ \Delta=-154,\ \det=-16,\ (n_+,n_-,n_0)=(1,7,0)$

[5] $21^{1}41\overset{1^{1}}{4}1^{1}$

$T=7,\ \Delta=-154,\ \det=-16,\ (n_+,n_-,n_0)=(1,7,0)$

[6] $21^{1}4141^{1}2$

$T=7,\ \Delta=-154,\ \det=-16,\ (n_+,n_-,n_0)=(1,7,0)$

$T=9$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{so}_8^3$ (6)

[1] $1\overset{1^{1}}{4}141412$

$T=9,\ \Delta=-124,\ \det=-1,\ (n_+,n_-,n_0)=(1,9,0)$

[2] $1\overset{1^{1}}{4}141\overset{1}{4}1$

$T=9,\ \Delta=-124,\ \det=-1,\ (n_+,n_-,n_0)=(1,9,0)$

[3] $1\overset{1^{1}}{4}141\overset{1^{1}}{4}1$

$T=9,\ \Delta=-124,\ \det=-4,\ (n_+,n_-,n_0)=(1,9,0)$

[4] $21^{1}4141\overset{1^{1}}{4}1^{1}$

$T=9,\ \Delta=-124,\ \det=-16,\ (n_+,n_-,n_0)=(1,9,0)$

[5] $1^{1}\overset{1^{1}}{4}141\overset{1^{1}}{4}1^{1}$

$T=9,\ \Delta=-124,\ \det=-16,\ (n_+,n_-,n_0)=(1,9,0)$

[6] $21^{1}414141^{1}2$

$T=9,\ \Delta=-124,\ \det=-16,\ (n_+,n_-,n_0)=(1,9,0)$

$T=11$, Ext $=\mathfrak{f}_4$, LST $=\mathfrak{e}_6\oplus\mathfrak{f}_4\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ (6)

[1] $1\overset{1^{1}3151321}{6}1$

$T=11,\ \Delta=-153,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[2] 21513216131^{1}

$T=11,\ \Delta=-153,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[3] $1\overset{1^{1}3161231}{5}1$

$T=11,\ \Delta=-153,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[4] 21513161231^{1}

$T=11,\ \Delta=-153,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[5] $1\overset{1^{1}3216131}{5}1$

$T=11,\ \Delta=-153,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[6] 21612315131^{1}

$T=11,\ \Delta=-153,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

$T=11$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{e}_6\oplus\mathfrak{f}_4\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ (6)

[1] $1\overset{1^{1}3151321}{6}1$

$T=11,\ \Delta=-129,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[2] 21513216131^{1}

$T=11,\ \Delta=-129,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[3] $1\overset{1^{1}3161231}{5}1$

$T=11,\ \Delta=-129,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[4] 21513161231^{1}

$T=11,\ \Delta=-129,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[5] $1\overset{1^{1}3216131}{5}1$

$T=11,\ \Delta=-129,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[6] 21612315131^{1}

$T=11,\ \Delta=-129,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

$T=11$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{so}_8^4$ (6)

[1] $1\overset{1^{1}}{4}14141\overset{1}{4}1$

$T=11,\ \Delta=-94,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[2] $1\overset{1^{1}}{4}14141412$

$T=11,\ \Delta=-94,\ \det=-1,\ (n_+,n_-,n_0)=(1,11,0)$

[3] $1\overset{1^{1}}{4}14141\overset{1^{1}}{4}1$

$T=11,\ \Delta=-94,\ \det=-4,\ (n_+,n_-,n_0)=(1,11,0)$

[4] $1^{1}\overset{1^{1}}{4}14141\overset{1^{1}}{4}1^{1}$

$T=11,\ \Delta=-94,\ \det=-16,\ (n_+,n_-,n_0)=(1,11,0)$

[5] $21^{1}414141\overset{1^{1}}{4}1^{1}$

$T=11,\ \Delta=-94,\ \det=-16,\ (n_+,n_-,n_0)=(1,11,0)$

[6] $21^{1}41414141^{1}2$

$T=11,\ \Delta=-94,\ \det=-16,\ (n_+,n_-,n_0)=(1,11,0)$

$T=12$, Ext $=\mathfrak{f}_4$, LST $=\mathfrak{e}_7\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2$ (6)

[1] $1\overset{1}{\underset{12321}{8}}1231^{1}$

$T=12,\ \Delta=-130,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[2] $1\,\overset{1}{\underset{1^{1}2321}{8}}\,1231$

$T=12,\ \Delta=-130,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[3] $21\,\overset{1^{1}2321}{8}\,1231$

$T=12,\ \Delta=-130,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[4] $21\,\overset{12321}{8}\,1231^{1}$

$T=12,\ \Delta=-130,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[5] $1\,\overset{1}{\underset{1^{1}2321}{8}}\,1231^{1}$

$T=12,\ \Delta=-130,\ \det=4,\ (n_+,n_-,n_0)=(1,12,0)$

[6] $21\,\overset{1^{1}2321}{8}\,1231^{1}$

$T=12,\ \Delta=-130,\ \det=4,\ (n_+,n_-,n_0)=(1,12,0)$

$T=12$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_7\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2$ **(6)**

[1] $1\,\overset{1}{\underset{1^{1}2321}{8}}\,1231$

$T=12,\ \Delta=-106,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[2] $1\,\overset{1}{\underset{12321}{8}}\,1231^{1}$

$T=12,\ \Delta=-106,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[3] $21\,\overset{1^{1}2321}{8}\,1231$

$T=12,\ \Delta=-106,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[4] $21\,\overset{12321}{8}\,1231^{1}$

$T=12,\ \Delta=-106,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[5] $1\,\overset{1}{\underset{1^{1}2321}{8}}\,1231^{1}$

$T=12,\ \Delta=-106,\ \det=4,\ (n_+,n_-,n_0)=(1,12,0)$

[6] $21\,\overset{1^{1}2321}{8}\,1231^{1}$

$T=12,\ \Delta=-106,\ \det=4,\ (n_+,n_-,n_0)=(1,12,0)$

$T=12$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_6\oplus\mathfrak{f}_4\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3^2$ **(6)**

[1] 131612315131^{1}

$T=12,\ \Delta=-108,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[2] 131513216131^{1}

$T=12,\ \Delta=-108,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[3] 132161315131^{1}

$T=12,\ \Delta=-108,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[4] 131513161231^{1}

$T=12,\ \Delta=-108,\ \det=1,\ (n_+,n_-,n_0)=(1,12,0)$

[5] $1^{1}31612315131^{1}$

$T=12,\ \Delta=-108,\ \det=4,\ (n_+,n_-,n_0)=(1,12,0)$

[6] $1^{1}32161315131^{1}$

$T=12,\ \Delta=-108,\ \det=4,\ (n_+,n_-,n_0)=(1,12,0)$

$T=13$**, Ext** $=\mathfrak{f}_4$**, LST** $=\mathfrak{e}_6\oplus\mathfrak{f}_4\oplus(\mathfrak{su}_2\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_3$ **(6)**

[1] 1316123151321^{1}

$T=13,\ \Delta=-104,\ \det=-1,\ (n_+,n_-,n_0)=(1,13,0)$

[2] 1231513216131^{1}

$T=13,\ \Delta=-104,\ \det=-1,\ (n_+,n_-,n_0)=(1,13,0)$

[3] 1321613151321^{1}

$T = 13,\ \Delta = -104,\ \det = -1,\ (n_+, n_-, n_0) = (1, 13, 0)$

[4] 1231513161231^{1}

$T = 13,\ \Delta = -104,\ \det = -1,\ (n_+, n_-, n_0) = (1, 13, 0)$

[5] $1^{1}316123151321^{1}$

$T = 13,\ \Delta = -104,\ \det = -4,\ (n_+, n_-, n_0) = (1, 13, 0)$

[6] $1^{1}321613151321^{1}$

$T = 13,\ \Delta = -104,\ \det = -4,\ (n_+, n_-, n_0) = (1, 13, 0)$

$T = 13$, Ext $= \mathfrak{so}_8$, LST $= \mathfrak{e}_6 \oplus \mathfrak{f}_4 \oplus (\mathfrak{su}_2 \oplus \mathfrak{g}_2)^2 \oplus \mathfrak{su}_3$ (6)

[1] 1316123151321^{1}

$T = 13,\ \Delta = -80,\ \det = -1,\ (n_+, n_-, n_0) = (1, 13, 0)$

[2] 1231513216131^{1}

$T = 13,\ \Delta = -80,\ \det = -1,\ (n_+, n_-, n_0) = (1, 13, 0)$

[3] 1231513161231^{1}

$T = 13,\ \Delta = -80,\ \det = -1,\ (n_+, n_-, n_0) = (1, 13, 0)$

[4] 1321613151321^{1}

$T = 13,\ \Delta = -80,\ \det = -1,\ (n_+, n_-, n_0) = (1, 13, 0)$

[5] $1^{1}316123151321^{1}$

$T = 13,\ \Delta = -80,\ \det = -4,\ (n_+, n_-, n_0) = (1, 13, 0)$

[6] $1^{1}321613151321^{1}$

$T = 13,\ \Delta = -80,\ \det = -4,\ (n_+, n_-, n_0) = (1, 13, 0)$

$T = 13$, Ext $= \mathfrak{so}_8$, LST $= \mathfrak{so}_8^5$ (6)

[1] $1\overset{1^{1}}{4}1414141412$

$T = 13,\ \Delta = -64,\ \det = -1,\ (n_+, n_-, n_0) = (1, 13, 0)$

[2] $1\overset{1^{1}}{4}1414141\overset{1}{4}1$

$T = 13,\ \Delta = -64,\ \det = -1,\ (n_+, n_-, n_0) = (1, 13, 0)$

[3] $1\overset{1^{1}}{4}1414141\overset{1^{1}}{4}1$

$T = 13,\ \Delta = -64,\ \det = -4,\ (n_+, n_-, n_0) = (1, 13, 0)$

[4] $21^{1}41414141\overset{1^{1}}{4}1^{1}$

$T = 13,\ \Delta = -64,\ \det = -16,\ (n_+, n_-, n_0) = (1, 13, 0)$

[5] $1^{1}\overset{1^{1}}{4}1414141\overset{1^{1}}{4}1^{1}$

$T = 13,\ \Delta = -64,\ \det = -16,\ (n_+, n_-, n_0) = (1, 13, 0)$

[6] $21^{1}4141414141^{1}2$

$T = 13,\ \Delta = -64,\ \det = -16,\ (n_+, n_-, n_0) = (1, 13, 0)$

$T = 14$, Ext $= \mathfrak{f}_4$, LST $= \mathfrak{e}_8 \oplus \mathfrak{sp}_1 \oplus \mathfrak{g}_2$ (10)

[1] $1\ \underset{1^{1}3221}{\overset{1}{(12)}}\ 122222$

$T = 14,\ \Delta = -176,\ \det = 1,\ (n_+, n_-, n_0) = (1, 14, 0)$

[2] $1\ \underset{1^{1}3221}{\overset{21}{(12)}}\ 12222$

$T = 14,\ \Delta = -176,\ \det = 1,\ (n_+, n_-, n_0) = (1, 14, 0)$

[3] $1\ \underset{1^{1}3221}{\overset{221}{(12)}}\ 1222$

$T = 14,\ \Delta = -176,\ \det = 1,\ (n_+, n_-, n_0) = (1, 14, 0)$

[4] $21\ \underset{1^{1}3221}{\overset{21}{(12)}}\ 1222$

$T=14,\ \Delta=-176,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[5] $21\ \overset{221}{\underset{1^{1}3221}{(12)}}\ 122$

$T=14,\ \Delta=-176,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[6] $1\ \overset{1^{1}3221}{(12)}\ 1222222$

$T=14,\ \Delta=-176,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[7] $21\ \overset{1^{1}3221}{(12)}\ 122222$

$T=14,\ \Delta=-176,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[8] $221\ \overset{1^{1}3221}{(12)}\ 12222$

$T=14,\ \Delta=-176,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[9] $2221\ \overset{1^{1}3221}{(12)}\ 1222$

$T=14,\ \Delta=-176,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[10] $22222221(12)12231^{1}$

$T=14,\ \Delta=-176,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

$T=14$, **Ext** $=\mathfrak{so}_8$, **LST** $=\mathfrak{e}_8\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2$ **(10)**

[1] $1\ \overset{1}{\underset{1^{1}3221}{(12)}}\ 122222$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[2] $1\ \overset{21}{\underset{1^{1}3221}{(12)}}\ 12222$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[3] $1\ \overset{221}{\underset{1^{1}3221}{(12)}}\ 1222$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[4] $21\ \overset{21}{\underset{1^{1}3221}{(12)}}\ 1222$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[5] $21\ \overset{221}{\underset{1^{1}3221}{(12)}}\ 122$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[6] $1\ \overset{1^{1}3221}{(12)}\ 1222222$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[7] $21\ \overset{1^{1}3221}{(12)}\ 122222$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[8] $221\ \overset{1^{1}3221}{(12)}\ 12222$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[9] $2221\ \overset{1^{1}3221}{(12)}\ 1222$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[10] $22222221(12)12231^{1}$

$T=14,\ \Delta=-152,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

$T=14$, **Ext** $=\mathfrak{f}_4$, **LST** $=\mathfrak{e}_6\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ **(6)**

[1] 13161231513221^{1}

$T=14,\ \Delta=-75,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[2] 12231513216131^{1}

$T=14,\ \Delta=-75,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[3] 12231513161231^{1}

$T=14,\ \Delta=-75,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[4] 13216131513221^{1}

$T=14,\ \Delta=-75,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[5] $1^{1}3161231513221^{1}$

$T=14,\ \Delta=-75,\ \det=4,\ (n_+,n_-,n_0)=(1,14,0)$

[6] $1^{1}3216131513221^{1}$

$T=14,\ \Delta=-75,\ \det=4,\ (n_+,n_-,n_0)=(1,14,0)$

$T=14$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_6\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ **(6)**

[1] 12231513216131^{1}

$T=14,\ \Delta=-51,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[2] 13161231513221^{1}

$T=14,\ \Delta=-51,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[3] 13216131513221^{1}

$T=14,\ \Delta=-51,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[4] 12231513161231^{1}

$T=14,\ \Delta=-51,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[5] $1^{1}3161231513221^{1}$

$T=14,\ \Delta=-51,\ \det=4,\ (n_+,n_-,n_0)=(1,14,0)$

[6] $1^{1}3216131513221^{1}$

$T=14,\ \Delta=-51,\ \det=4,\ (n_+,n_-,n_0)=(1,14,0)$

$T=14$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_7\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ **(6)**

[1] 13151322181231^{1}

$T=14,\ \Delta=-106,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[2] 13218122315131^{1}

$T=14,\ \Delta=-106,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[3] 13151321812231^{1}

$T=14,\ \Delta=-106,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[4] 13221812315131^{1}

$T=14,\ \Delta=-106,\ \det=1,\ (n_+,n_-,n_0)=(1,14,0)$

[5] $1^{1}3218122315131^{1}$

$T=14,\ \Delta=-106,\ \det=4,\ (n_+,n_-,n_0)=(1,14,0)$

[6] $1^{1}3221812315131^{1}$

$T=14,\ \Delta=-106,\ \det=4,\ (n_+,n_-,n_0)=(1,14,0)$

$T=15$**, Ext** $=\mathfrak{f}_4$**, LST** $=\mathfrak{e}_8\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2$ **(6)**

[1] $1\underset{13221}{\overset{221}{(12)}}12231^{1}$

$T=15,\ \Delta=-156,\ \det=-1,\ (n_+,n_-,n_0)=(1,15,0)$

[2] $21\underset{13221}{\overset{21}{(12)}}12231^{1}$

$T=15,\ \Delta=-156,\ \det=-1,\ (n_+,n_-,n_0)=(1,15,0)$

[3] $2221\overset{13221}{(12)}12231^{1}$

$T=15,\ \Delta=-156,\ \det=-1,\ (n_+,n_-,n_0)=(1,15,0)$

[4] $1\ \underset{1^{1}3221}{\overset{221}{(12)}}\ 12231^{1}$

$T=15,\ \Delta=-156,\ \det=-4,\ (n_+,n_-,n_0)=(1,15,0)$

[5] $21\ \underset{1^{1}3221}{\overset{21}{(12)}}\ 12231^{1}$

$T = 15,\ \Delta = -156,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

[6] $2221\,\overset{1^{1}3221}{(12)}\,12231^{1}$

$T = 15,\ \Delta = -156,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

$T = 15$, Ext $= \mathfrak{f}_4$, LST $= \mathfrak{e}_7 \oplus \mathfrak{f}_4 \oplus \mathfrak{sp}_1 \oplus \mathfrak{g}_2 \oplus (\mathfrak{su}_2 \oplus \mathfrak{g}_2)^2$ (6)

[1] 132181223151321^{1}

$T = 15,\ \Delta = -102,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[2] 123151322181231^{1}

$T = 15,\ \Delta = -102,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[3] 123151321812231^{1}

$T = 15,\ \Delta = -102,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[4] 132218123151321^{1}

$T = 15,\ \Delta = -102,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[5] $1^{1}32181223151321^{1}$

$T = 15,\ \Delta = -102,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

[6] $1^{1}32218123151321^{1}$

$T = 15,\ \Delta = -102,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

$T = 15$, Ext $= \mathfrak{so}_8$, LST $= \mathfrak{e}_8 \oplus (\mathfrak{sp}_1 \oplus \mathfrak{g}_2)^2$ (6)

[1] $1\underset{13221}{\overset{221}{(12)}}12231^{1}$

$T = 15,\ \Delta = -132,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[2] $21\underset{13221}{\overset{21}{(12)}}12231^{1}$

$T = 15,\ \Delta = -132,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[3] $2221\overset{13221}{(12)}12231^{1}$

$T = 15,\ \Delta = -132,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[4] $1\,\underset{1^{1}3221}{\overset{221}{(12)}}\,12231^{1}$

$T = 15,\ \Delta = -132,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

[5] $21\,\underset{1^{1}3221}{\overset{21}{(12)}}\,12231^{1}$

$T = 15,\ \Delta = -132,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

[6] $2221\,\overset{1^{1}3221}{(12)}\,12231^{1}$

$T = 15,\ \Delta = -132,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

$T = 15$, Ext $= \mathfrak{so}_8$, LST $= \mathfrak{e}_7 \oplus \mathfrak{f}_4 \oplus \mathfrak{sp}_1 \oplus \mathfrak{g}_2 \oplus (\mathfrak{su}_2 \oplus \mathfrak{g}_2)^2$ (6)

[1] 132181223151321^{1}

$T = 15,\ \Delta = -78,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[2] 123151322181231^{1}

$T = 15,\ \Delta = -78,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[3] 132218123151321^{1}

$T = 15,\ \Delta = -78,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[4] 123151321812231^{1}

$T = 15,\ \Delta = -78,\ \det = -1,\ (n_+, n_-, n_0) = (1, 15, 0)$

[5] $1^{1}32181223151321^{1}$

$T = 15,\ \Delta = -78,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

[6] $1^{1}32218123151321^{1}$

$T = 15,\ \Delta = -78,\ \det = -4,\ (n_+, n_-, n_0) = (1, 15, 0)$

$T=15$**,** **Ext** $=\mathfrak{so}_8$**,** **LST** $=\mathfrak{so}_8^6$ **(6)**

[1] $1\overset{1^{1}}{4}141414141\overset{1}{4}1$

$T=15,\ \Delta=-34,\ \det=-1,\ (n_+,n_-,n_0)=(1,15,0)$

[2] $1\overset{1^{1}}{4}141414141412$

$T=15,\ \Delta=-34,\ \det=-1,\ (n_+,n_-,n_0)=(1,15,0)$

[3] $1\overset{1^{1}}{4}141414141\overset{1^{1}}{4}1$

$T=15,\ \Delta=-34,\ \det=-4,\ (n_+,n_-,n_0)=(1,15,0)$

[4] $1^{1}\overset{1^{1}}{4}141414141\overset{1^{1}}{4}1^{1}$

$T=15,\ \Delta=-34,\ \det=-16,\ (n_+,n_-,n_0)=(1,15,0)$

[5] $21^{1}4141414141\overset{1^{1}}{4}1^{1}$

$T=15,\ \Delta=-34,\ \det=-16,\ (n_+,n_-,n_0)=(1,15,0)$

[6] $21^{1}414141414141^{1}2$

$T=15,\ \Delta=-34,\ \det=-16,\ (n_+,n_-,n_0)=(1,15,0)$

$T=16$**,** **Ext** $=\mathfrak{f}_4$**,** **LST** $=\mathfrak{e}_7\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ **(6)**

[1] 1321812231513221^{1}

$T=16,\ \Delta=-73,\ \det=1,\ (n_+,n_-,n_0)=(1,16,0)$

[2] 1223151322181231^{1}

$T=16,\ \Delta=-73,\ \det=1,\ (n_+,n_-,n_0)=(1,16,0)$

[3] 1223151321812231^{1}

$T=16,\ \Delta=-73,\ \det=1,\ (n_+,n_-,n_0)=(1,16,0)$

[4] 1322181231513221^{1}

$T=16,\ \Delta=-73,\ \det=1,\ (n_+,n_-,n_0)=(1,16,0)$

[5] $1^{1}321812231513221^{1}$

$T=16,\ \Delta=-73,\ \det=4,\ (n_+,n_-,n_0)=(1,16,0)$

[6] $1^{1}322181231513221^{1}$

$T=16,\ \Delta=-73,\ \det=4,\ (n_+,n_-,n_0)=(1,16,0)$

$T=16$**,** **Ext** $=\mathfrak{so}_8$**,** **LST** $=\mathfrak{e}_7\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ **(6)**

[1] 1223151322181231^{1}

$T=16,\ \Delta=-49,\ \det=1,\ (n_+,n_-,n_0)=(1,16,0)$

[2] 1321812231513221^{1}

$T=16,\ \Delta=-49,\ \det=1,\ (n_+,n_-,n_0)=(1,16,0)$

[3] 1223151321812231^{1}

$T=16,\ \Delta=-49,\ \det=1,\ (n_+,n_-,n_0)=(1,16,0)$

[4] 1322181231513221^{1}

$T=16,\ \Delta=-49,\ \det=1,\ (n_+,n_-,n_0)=(1,16,0)$

[5] $1^{1}321812231513221^{1}$

$T=16,\ \Delta=-49,\ \det=4,\ (n_+,n_-,n_0)=(1,16,0)$

[6] $1^{1}322181231513221^{1}$

$T=16,\ \Delta=-49,\ \det=4,\ (n_+,n_-,n_0)=(1,16,0)$

$T=17$**,** **Ext** $=\mathfrak{e}_6$**,** **LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ **(8)**

[1] $1\quad\underset{1^{1}31513221}{\overset{1}{(12)}}\quad 12222$

$T=17,\ \Delta=-175,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[2] $1\quad\underset{1^{1}31513221}{\overset{21}{(12)}}\quad 1222$

$T = 17,\ \Delta = -175,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[3] $1 \underset{1^{1}31513221}{\overset{221}{(12)}} 122$

$T = 17,\ \Delta = -175,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[4] $21 \underset{1^{1}31513221}{\overset{221}{(12)}} 12$

$T = 17,\ \Delta = -175,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[5] $1 \overset{1^{1}31513221}{(12)} 122222$

$T = 17,\ \Delta = -175,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[6] $21 \overset{1^{1}31513221}{(12)} 12222$

$T = 17,\ \Delta = -175,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[7] $221 \overset{1^{1}31513221}{(12)} 1222$

$T = 17,\ \Delta = -175,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[8] $2222221(12)122315131^{1}$

$T = 17,\ \Delta = -175,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

$T = 17$, **Ext** $= \mathfrak{f}_4$, **LST** $= \mathfrak{e}_8 \oplus \mathfrak{f}_4 \oplus \mathfrak{sp}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{su}_3$ **(8)**

[1] $1 \underset{1^{1}31513221}{\overset{1}{(12)}} 12222$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[2] $1 \underset{1^{1}31513221}{\overset{21}{(12)}} 1222$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[3] $1 \underset{1^{1}31513221}{\overset{221}{(12)}} 122$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[4] $21 \underset{1^{1}31513221}{\overset{221}{(12)}} 12$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[5] $1 \overset{1^{1}31513221}{(12)} 122222$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[6] $21 \overset{1^{1}31513221}{(12)} 12222$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[7] $221 \overset{1^{1}31513221}{(12)} 1222$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[8] $2222221(12)122315131^{1}$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

$T = 17$, **Ext** $= \mathfrak{so}_8$, **LST** $= \mathfrak{e}_8 \oplus \mathfrak{f}_4 \oplus \mathfrak{sp}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{su}_3$ **(8)**

[1] $1 \underset{1^{1}31513221}{\overset{1}{(12)}} 12222$

$T = 17,\ \Delta = -125,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[2] $1 \underset{1^{1}31513221}{\overset{21}{(12)}} 1222$

$T = 17,\ \Delta = -125,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[3] $1 \underset{1^{1}31513221}{\overset{221}{(12)}} 122$

$T = 17,\ \Delta = -125,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[4] $21\overset{221}{\underset{1^{1}31513221}{(12)}}12$

$T = 17,\ \Delta = -125,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[5] $1\overset{1^{1}31513221}{(12)}122222$

$T = 17,\ \Delta = -125,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[6] $21\overset{1^{1}31513221}{(12)}12222$

$T = 17,\ \Delta = -125,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[7] $221\overset{1^{1}31513221}{(12)}1222$

$T = 17,\ \Delta = -125,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[8] $2222221(12)122315131^{1}$

$T = 17,\ \Delta = -125,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

$T = 17$, **Ext** $= \mathfrak{e}_6$, **LST** $= \mathfrak{e}_6 \oplus \mathfrak{e}_7 \oplus \mathfrak{f}_4 \oplus \mathfrak{su}_2 \oplus \mathfrak{g}_2 \oplus \mathfrak{su}_2 \oplus \mathfrak{so}_7 \oplus \mathfrak{su}_2 \oplus \mathfrak{su}_3$ **(6)**

[1] $1\overset{1}{6}12321812315131^{1}$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[2] $1\overset{1}{6}13151321812321^{1}$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[3] 21513218123216131^{1}

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[4] $1\overset{1^{1}3161232181231}{5}1$

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[5] 21612321812315131^{1}

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[6] 21613151321812321^{1}

$T = 17,\ \Delta = -149,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

$T = 17$, **Ext** $= \mathfrak{f}_4$, **LST** $= \mathfrak{e}_6 \oplus \mathfrak{e}_7 \oplus \mathfrak{f}_4 \oplus \mathfrak{su}_2 \oplus \mathfrak{g}_2 \oplus \mathfrak{su}_2 \oplus \mathfrak{so}_7 \oplus \mathfrak{su}_2 \oplus \mathfrak{su}_3$ **(6)**

[1] $1\overset{1}{6}12321812315131^{1}$

$T = 17,\ \Delta = -123,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[2] $1\overset{1}{6}13151321812321^{1}$

$T = 17,\ \Delta = -123,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[3] 21513218123216131^{1}

$T = 17,\ \Delta = -123,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[4] $1\overset{1^{1}3161232181231}{5}1$

$T = 17,\ \Delta = -123,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[5] 21612321812315131^{1}

$T = 17,\ \Delta = -123,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[6] 21613151321812321^{1}

$T = 17,\ \Delta = -123,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

$T = 17$, **Ext** $= \mathfrak{f}_4$, **LST** $= \mathfrak{e}_7 \oplus \mathfrak{e}_7' \oplus \mathfrak{su}_2 \oplus \mathfrak{g}_2 \oplus (\mathfrak{su}_2 \oplus \mathfrak{so}_7 \oplus \mathfrak{su}_2)^2$ **(6)**

[1] $1\overset{1^{1}321}{8}12321712321$

$T = 17,\ \Delta = -101,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[2] $1\overset{1321}{8}12321712321^{1}$

$T = 17,\ \Delta = -101,\ \det = -1,\ (n_+, n_-, n_0) = (1, 17, 0)$

[3] $1\overset{1^{1}2321}{8}1232171231$

$T=17,\ \Delta=-101,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[4] $1\overset{12321}{8}1232171231^{1}$

$T=17,\ \Delta=-101,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[5] $1\overset{1^{1}321}{8}12321712321^{1}$

$T=17,\ \Delta=-101,\ \det=-4,\ (n_+,n_-,n_0)=(1,17,0)$

[6] $1\overset{1^{1}2321}{8}1232171231^{1}$

$T=17,\ \Delta=-101,\ \det=-4,\ (n_+,n_-,n_0)=(1,17,0)$

$T=17$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{so}_8^7$ (6)

[1] $1\overset{1^{1}}{4}1414141414141\overset{1}{4}1$

$T=17,\ \Delta=-4,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[2] $1\overset{1^{1}}{4}1414141414141412$

$T=17,\ \Delta=-4,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[3] $1\overset{1^{1}}{4}1414141414141\overset{1^{1}}{4}1$

$T=17,\ \Delta=-4,\ \det=-4,\ (n_+,n_-,n_0)=(1,17,0)$

[4] $1^{1}\overset{1^{1}}{4}1414141414141\overset{1^{1}}{4}1^{1}$

$T=17,\ \Delta=-4,\ \det=-16,\ (n_+,n_-,n_0)=(1,17,0)$

[5] $21^{1}41414141414141\overset{1^{1}}{4}1^{1}$

$T=17,\ \Delta=-4,\ \det=-16,\ (n_+,n_-,n_0)=(1,17,0)$

[6] $21^{1}4141414141414141^{1}2$

$T=17,\ \Delta=-4,\ \det=-16,\ (n_+,n_-,n_0)=(1,17,0)$

$T=17$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{e}_6\oplus\mathfrak{e}_7\oplus\mathfrak{f}_4\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2\oplus\mathfrak{su}_3$ (6)

[1] $1\overset{1}{6}12321812315131^{1}$

$T=17,\ \Delta=-99,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[2] $1\overset{1}{6}13151321812321^{1}$

$T=17,\ \Delta=-99,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[3] 21513218123216131^{1}

$T=17,\ \Delta=-99,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[4] $1\overset{1^{1}3161232181231}{5}1$

$T=17,\ \Delta=-99,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[5] 21612321812315131^{1}

$T=17,\ \Delta=-99,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[6] 21613151321812321^{1}

$T=17,\ \Delta=-99,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

$T=17$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{e}_7\oplus\mathfrak{e}_7'\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus(\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2)^2$ (6)

[1] $1\overset{1321}{8}12321712321^{1}$

$T=17,\ \Delta=-77,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[2] $1\overset{1^{1}321}{8}12321712321$

$T=17,\ \Delta=-77,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[3] $1\overset{12321}{8}1232171231^{1}$

$T=17,\ \Delta=-77,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[4] $1\overset{1^{1}2321}{8}1232171231$

$T=17,\ \Delta=-77,\ \det=-1,\ (n_+,n_-,n_0)=(1,17,0)$

[5] $1\overset{1^{1}321}{8}12321712321^{1}$

$T=17,\ \Delta=-77,\ \det=-4,\ (n_+,n_-,n_0)=(1,17,0)$

[6] $1\overset{1^{1}2321}{8}1232171231^{1}$

$T=17,\ \Delta=-77,\ \det=-4,\ (n_+,n_-,n_0)=(1,17,0)$

$T=18$, Ext $=\mathfrak{e}_6$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ (8)

[1] $1\underset{1^{1}231513221}{\overset{1}{(12)}}12222$

$T=18,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] $1\underset{1^{1}231513221}{\overset{21}{(12)}}1222$

$T=18,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] $1\underset{1^{1}231513221}{\overset{221}{(12)}}122$

$T=18,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] $21\underset{1^{1}231513221}{\overset{221}{(12)}}12$

$T=18,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1\overset{1^{1}231513221}{(12)}122222$

$T=18,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $21\overset{1^{1}231513221}{(12)}12222$

$T=18,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[7] $221\overset{1^{1}231513221}{(12)}1222$

$T=18,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[8] $2222221(12)1223151321^{1}$

$T=18,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

$T=18$, Ext $=\mathfrak{e}_7$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ (8)

[1] $1\underset{1^{1}231513221}{\overset{1}{(12)}}12222$

$T=18,\ \Delta=-202,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] $1\underset{1^{1}231513221}{\overset{21}{(12)}}1222$

$T=18,\ \Delta=-202,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] $1\underset{1^{1}231513221}{\overset{221}{(12)}}122$

$T=18,\ \Delta=-202,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] $21\underset{1^{1}231513221}{\overset{221}{(12)}}12$

$T=18,\ \Delta=-202,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1\overset{1^{1}231513221}{(12)}122222$

$T=18,\ \Delta=-202,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $21\overset{1^{1}231513221}{(12)}12222$

$T=18,\ \Delta=-202,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[7] $221\overset{1^{1}231513221}{(12)}1222$

$T=18,\ \Delta=-202,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[8] $2222221(12)1223151321^{1}$

$T=18,\ \Delta=-202,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

$T=18$, Ext $=\mathfrak{e}_7'$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ (8)

[1] $1\ \ \overset{1}{\underset{1^{1}231513221}{(12)}}\ \ 12222$

$T=18,\ \Delta=-174,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] $1\ \ \overset{21}{\underset{1^{1}231513221}{(12)}}\ \ 1222$

$T=18,\ \Delta=-174,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] $1\ \ \overset{221}{\underset{1^{1}231513221}{(12)}}\ \ 122$

$T=18,\ \Delta=-174,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] $21\ \ \overset{221}{\underset{1^{1}231513221}{(12)}}\ \ 12$

$T=18,\ \Delta=-174,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1\ \ \overset{1^{1}231513221}{(12)}\ \ 122222$

$T=18,\ \Delta=-174,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $21\ \ \overset{1^{1}231513221}{(12)}\ \ 12222$

$T=18,\ \Delta=-174,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[7] $221\ \ \overset{1^{1}231513221}{(12)}\ \ 1222$

$T=18,\ \Delta=-174,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[8] $2222221(12)1223151321^{1}$

$T=18,\ \Delta=-174,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

$T=18$, Ext $=\mathfrak{f}_4$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ (8)

[1] $1\ \ \overset{1}{\underset{1^{1}231513221}{(12)}}\ \ 12222$

$T=18,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] $1\ \ \overset{21}{\underset{1^{1}231513221}{(12)}}\ \ 1222$

$T=18,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] $1\ \ \overset{221}{\underset{1^{1}231513221}{(12)}}\ \ 122$

$T=18,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] $21\ \ \overset{221}{\underset{1^{1}231513221}{(12)}}\ \ 12$

$T=18,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1\ \ \overset{1^{1}231513221}{(12)}\ \ 122222$

$T=18,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $21\ \ \overset{1^{1}231513221}{(12)}\ \ 12222$

$T=18,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[7] $221\ \ \overset{1^{1}231513221}{(12)}\ \ 1222$

$T=18,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[8] $2222221(12)1223151321^{1}$

$T=18,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

$T=18$, **Ext** $=\mathfrak{so}_8$, **LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus\mathfrak{sp}_1\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ **(8)**

[1] $1\ \ \overset{1}{\underset{1^{1}231513221}{(12)}}\ \ 12222$

$T=18,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] $1\ \ \overset{21}{\underset{1^{1}231513221}{(12)}}\ \ 1222$

$T=18,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] $1\ \ \overset{221}{\underset{1^{1}231513221}{(12)}}\ \ 122$

$T=18,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] $21\ \ \overset{221}{\underset{1^{1}231513221}{(12)}}\ \ 12$

$T=18,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1\ \ \overset{1^{1}231513221}{(12)}\ \ 122222$

$T=18,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $21\ \ \overset{1^{1}231513221}{(12)}\ \ 12222$

$T=18,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[7] $221\ \ \overset{1^{1}231513221}{(12)}\ \ 1222$

$T=18,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[8] $2222221(12)1223151321^{1}$

$T=18,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

$T=18$, **Ext** $=\mathfrak{f}_4$, **LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_3$ **(6)**

[1] $1\ \ \overset{21}{\underset{131513221}{(12)}}\ \ 12231^{1}$

$T=18,\ \Delta=-129,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] $1\ \ \overset{21}{\underset{1^{1}31513221}{(12)}}\ \ 12231$

$T=18,\ \Delta=-129,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] $221\ \ \overset{131513221}{(12)}\ \ 12231^{1}$

$T=18,\ \Delta=-129,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] $221\ \ \overset{1^{1}31513221}{(12)}\ \ 12231$

$T=18,\ \Delta=-129,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1\ \ \overset{21}{\underset{1^{1}31513221}{(12)}}\ \ 12231^{1}$

$T=18,\ \Delta=-129,\ \det=4,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $221\ \ \overset{1^{1}31513221}{(12)}\ \ 12231^{1}$

$T=18,\ \Delta=-129,\ \det=4,\ (n_+,n_-,n_0)=(1,18,0)$

$T=18$, **Ext** $=\mathfrak{f}_4$, **LST** $=\mathfrak{e}_6\oplus\mathfrak{e}_7\oplus\mathfrak{f}_4\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2\oplus\mathfrak{su}_3^2$ **(6)**

[1] $1316123218123151 31^{1}$

$T=18,\ \Delta=-102,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] 131513218123216131^{1}

$T=18,\ \Delta=-102,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] 131613151321812321^{1}

$T=18,\ \Delta=-102,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] 123218123151316131^{1}

$T=18,\ \Delta=-102,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1^{1}31612321812315131^{1}$

$T=18,\ \Delta=-102,\ \det=4,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $1^{1}31613151321812321^{1}$

$T=18,\ \Delta=-102,\ \det=4,\ (n_+,n_-,n_0)=(1,18,0)$

$T=18$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_3$ **(6)**

[1] $1\ \underset{1^{1}31513221}{\overset{21}{(12)}}\ 12231$

$T=18,\ \Delta=-105,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] $1\ \underset{131513221}{\overset{21}{(12)}}\ 12231^{1}$

$T=18,\ \Delta=-105,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] $221\ \overset{1^{1}31513221}{(12)}\ 12231$

$T=18,\ \Delta=-105,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] $221\ \overset{131513221}{(12)}\ 12231^{1}$

$T=18,\ \Delta=-105,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1\ \underset{1^{1}31513221}{\overset{21}{(12)}}\ 12231^{1}$

$T=18,\ \Delta=-105,\ \det=4,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $221\ \overset{1^{1}31513221}{(12)}\ 12231^{1}$

$T=18,\ \Delta=-105,\ \det=4,\ (n_+,n_-,n_0)=(1,18,0)$

$T=18$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_6\oplus\mathfrak{e}_7\oplus\mathfrak{f}_4\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2\oplus\mathfrak{su}_3^2$ **(6)**

[1] 131612321812315131^{1}

$T=18,\ \Delta=-78,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[2] 131513218123216131^{1}

$T=18,\ \Delta=-78,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[3] 123218123151316131^{1}

$T=18,\ \Delta=-78,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[4] 131613151321812321^{1}

$T=18,\ \Delta=-78,\ \det=1,\ (n_+,n_-,n_0)=(1,18,0)$

[5] $1^{1}31612321812315131^{1}$

$T=18,\ \Delta=-78,\ \det=4,\ (n_+,n_-,n_0)=(1,18,0)$

[6] $1^{1}31613151321812321^{1}$

$T=18,\ \Delta=-78,\ \det=4,\ (n_+,n_-,n_0)=(1,18,0)$

$T=19$**, Ext** $=\mathfrak{e}_6$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2$ **(8)**

[1] $1\ \underset{1^{1}2231513221}{\overset{1}{(12)}}\ 12222$

$T=19,\ \Delta=-118,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[2] $1\ \underset{1^{1}2231513221}{\overset{21}{(12)}}\ 1222$

$T=19,\ \Delta=-118,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[3] $1\ \underset{1^{1}2231513221}{\overset{221}{(12)}}\ 122$

$T=19,\ \Delta=-118,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[4] $21\ \underset{1^{1}2231513221}{\overset{221}{(12)}}\ 12$

$T=19,\ \Delta=-118,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[5] $1\overset{1^{1}2231513221}{(12)}122222$

$T=19,\ \Delta=-118,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[6] $21\overset{1^{1}2231513221}{(12)}12222$

$T=19,\ \Delta=-118,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[7] $221\overset{1^{1}2231513221}{(12)}1222$

$T=19,\ \Delta=-118,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[8] $2222221(12)12231513221^{1}$

$T=19,\ \Delta=-118,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

$T=19$**, Ext** $=\mathfrak{e}_7$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2$ **(8)**

[1] $1\underset{1^{1}2231513221}{\overset{1}{(12)}}12222$

$T=19,\ \Delta=-173,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[2] $1\underset{1^{1}2231513221}{\overset{21}{(12)}}1222$

$T=19,\ \Delta=-173,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[3] $1\underset{1^{1}2231513221}{\overset{221}{(12)}}122$

$T=19,\ \Delta=-173,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[4] $21\underset{1^{1}2231513221}{\overset{221}{(12)}}12$

$T=19,\ \Delta=-173,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[5] $1\overset{1^{1}2231513221}{(12)}122222$

$T=19,\ \Delta=-173,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[6] $21\overset{1^{1}2231513221}{(12)}12222$

$T=19,\ \Delta=-173,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[7] $221\overset{1^{1}2231513221}{(12)}1222$

$T=19,\ \Delta=-173,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[8] $2222221(12)12231513221^{1}$

$T=19,\ \Delta=-173,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

$T=19$**, Ext** $=\mathfrak{e}_7'$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2$ **(8)**

[1] $1\underset{1^{1}2231513221}{\overset{1}{(12)}}12222$

$T=19,\ \Delta=-145,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[2] $1\underset{1^{1}2231513221}{\overset{21}{(12)}}1222$

$T=19,\ \Delta=-145,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[3] $1\underset{1^{1}2231513221}{\overset{221}{(12)}}122$

$T=19,\ \Delta=-145,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[4] $21\underset{1^{1}2231513221}{\overset{221}{(12)}}12$

$T=19,\ \Delta=-145,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[5] $1\overset{1^{1}2231513221}{(12)}122222$

$T=19,\ \Delta=-145,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[6] $21\overset{1^{1}2231513221}{(12)}12222$

$T = 19,\ \Delta = -145,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[7] $221\overset{1^{1}2231513221}{(12)}1222$

$T = 19,\ \Delta = -145,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[8] $2222221(12)12231513221^{1}$

$T = 19,\ \Delta = -145,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

$T = 19$**, Ext** $= \mathfrak{e}_8$**, LST** $= \mathfrak{e}_8 \oplus \mathfrak{f}_4 \oplus (\mathfrak{sp}_1 \oplus \mathfrak{g}_2)^2$ **(8)**

[1] $1\overset{1}{\underset{1^{1}2231513221}{(12)}}12222$

$T = 19,\ \Delta = -288,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[2] $1\overset{21}{\underset{1^{1}2231513221}{(12)}}1222$

$T = 19,\ \Delta = -288,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[3] $1\overset{221}{\underset{1^{1}2231513221}{(12)}}122$

$T = 19,\ \Delta = -288,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[4] $21\overset{221}{\underset{1^{1}2231513221}{(12)}}12$

$T = 19,\ \Delta = -288,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[5] $1\overset{1^{1}2231513221}{(12)}122222$

$T = 19,\ \Delta = -288,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[6] $21\overset{1^{1}2231513221}{(12)}12222$

$T = 19,\ \Delta = -288,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[7] $221\overset{1^{1}2231513221}{(12)}1222$

$T = 19,\ \Delta = -288,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[8] $2222221(12)12231513221^{1}$

$T = 19,\ \Delta = -288,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

$T = 19$**, Ext** $= \mathfrak{f}_4$**, LST** $= \mathfrak{e}_8 \oplus \mathfrak{f}_4 \oplus (\mathfrak{sp}_1 \oplus \mathfrak{g}_2)^2$ **(8)**

[1] $1\overset{1}{\underset{1^{1}2231513221}{(12)}}12222$

$T = 19,\ \Delta = -92,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[2] $1\overset{21}{\underset{1^{1}2231513221}{(12)}}1222$

$T = 19,\ \Delta = -92,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[3] $1\overset{221}{\underset{1^{1}2231513221}{(12)}}122$

$T = 19,\ \Delta = -92,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[4] $21\overset{221}{\underset{1^{1}2231513221}{(12)}}12$

$T = 19,\ \Delta = -92,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[5] $1\overset{1^{1}2231513221}{(12)}122222$

$T = 19,\ \Delta = -92,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[6] $21\overset{1^{1}2231513221}{(12)}12222$

$T = 19,\ \Delta = -92,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[7] $221\overset{1^{1}2231513221}{(12)}1222$

$T = 19,\ \Delta = -92,\ \det = -1,\ (n_+, n_-, n_0) = (1, 19, 0)$

[8] $2222221(12)12231513221^{1}$

$T=19,\ \Delta=-92,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

$T=19$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2$ **(8)**

[1] $1\ \underset{1^{1}2231513221}{\overset{1}{(12)}}\ 12222$

$T=19,\ \Delta=-68,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[2] $1\ \underset{1^{1}2231513221}{\overset{21}{(12)}}\ 1222$

$T=19,\ \Delta=-68,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[3] $1\ \underset{1^{1}2231513221}{\overset{221}{(12)}}\ 122$

$T=19,\ \Delta=-68,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[4] $21\ \underset{1^{1}2231513221}{\overset{221}{(12)}}\ 12$

$T=19,\ \Delta=-68,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[5] $1\ \overset{1^{1}2231513221}{(12)}\ 122222$

$T=19,\ \Delta=-68,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[6] $21\ \overset{1^{1}2231513221}{(12)}\ 12222$

$T=19,\ \Delta=-68,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[7] $221\ \overset{1^{1}2231513221}{(12)}\ 1222$

$T=19,\ \Delta=-68,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[8] $2222221(12)12231513221^{1}$

$T=19,\ \Delta=-68,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

$T=19$**, Ext** $=\mathfrak{f}_4$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ **(6)**

[1] $1\ \underset{1231513221}{\overset{21}{(12)}}\ 12231^{1}$

$T=19,\ \Delta=-101,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[2] $1\ \underset{1^{1}231513221}{\overset{21}{(12)}}\ 12231$

$T=19,\ \Delta=-101,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[3] $221\ \overset{1^{1}231513221}{(12)}\ 12231$

$T=19,\ \Delta=-101,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[4] $221\ \overset{1231513221}{(12)}\ 12231^{1}$

$T=19,\ \Delta=-101,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[5] $1\ \underset{1^{1}231513221}{\overset{21}{(12)}}\ 12231^{1}$

$T=19,\ \Delta=-101,\ \det=-4,\ (n_+,n_-,n_0)=(1,19,0)$

[6] $221\ \overset{1^{1}231513221}{(12)}\ 12231^{1}$

$T=19,\ \Delta=-101,\ \det=-4,\ (n_+,n_-,n_0)=(1,19,0)$

$T=19$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ **(6)**

[1] $1\ \underset{1231513221}{\overset{21}{(12)}}\ 12231^{1}$

$T=19,\ \Delta=-77,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[2] $1\ \underset{1^{1}231513221}{\overset{21}{(12)}}\ 12231$

$T=19,\ \Delta=-77,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[3] $221\overset{1231513221}{(12)}12231^{1}$

$T=19,\ \Delta=-77,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[4] $221\overset{1^{1}231513221}{(12)}12231$

$T=19,\ \Delta=-77,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[5] $1\underset{1^{1}231513221}{\overset{21}{(12)}}12231^{1}$

$T=19,\ \Delta=-77,\ \det=-4,\ (n_+,n_-,n_0)=(1,19,0)$

[6] $221\overset{1^{1}231513221}{(12)}12231^{1}$

$T=19,\ \Delta=-77,\ \det=-4,\ (n_+,n_-,n_0)=(1,19,0)$

$T=19$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{so}_8^4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2$ **(6)**

[1] $1414141413221(12)12231^{1}$

$T=19,\ \Delta=-128,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[2] $13221(12)1223141414141^{1}$

$T=19,\ \Delta=-128,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[3] $1413221(12)12231414141^{1}$

$T=19,\ \Delta=-128,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[4] $14141413221(12)1223141^{1}$

$T=19,\ \Delta=-128,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[5] $141413221(12)122314141^{1}$

$T=19,\ \Delta=-128,\ \det=-1,\ (n_+,n_-,n_0)=(1,19,0)$

[6] $1^{1}414141413221(12)12231^{1}$

$T=19,\ \Delta=-128,\ \det=-4,\ (n_+,n_-,n_0)=(1,19,0)$

$T=20$**, Ext** $=\mathfrak{f}_4$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3$ **(6)**

[1] $1\underset{12231513221}{\overset{21}{(12)}}12231^{1}$

$T=20,\ \Delta=-72,\ \det=1,\ (n_+,n_-,n_0)=(1,20,0)$

[2] $1\underset{1^{1}2231513221}{\overset{21}{(12)}}12231$

$T=20,\ \Delta=-72,\ \det=1,\ (n_+,n_-,n_0)=(1,20,0)$

[3] $221\overset{1^{1}2231513221}{(12)}12231$

$T=20,\ \Delta=-72,\ \det=1,\ (n_+,n_-,n_0)=(1,20,0)$

[4] $221\overset{12231513221}{(12)}12231^{1}$

$T=20,\ \Delta=-72,\ \det=1,\ (n_+,n_-,n_0)=(1,20,0)$

[5] $1\underset{1^{1}2231513221}{\overset{21}{(12)}}12231^{1}$

$T=20,\ \Delta=-72,\ \det=4,\ (n_+,n_-,n_0)=(1,20,0)$

[6] $221\overset{1^{1}2231513221}{(12)}12231^{1}$

$T=20,\ \Delta=-72,\ \det=4,\ (n_+,n_-,n_0)=(1,20,0)$

$T=20$**, Ext** $=\mathfrak{so}_8$**, LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3$ **(6)**

[1] $1\underset{1^{1}2231513221}{\overset{21}{(12)}}12231$

$T=20,\ \Delta=-48,\ \det=1,\ (n_+,n_-,n_0)=(1,20,0)$

[2] $1\underset{12231513221}{\overset{21}{(12)}}12231^{1}$

$T=20,\ \Delta=-48,\ \det=1,\ (n_+,n_-,n_0)=(1,20,0)$

[3] $221 \quad \overset{1^{1}2231513221}{(12)} \quad 12231$

$T=20,\ \Delta=-48,\ \det=1,\ (n_+,n_-,n_0)=(1,20,0)$

[4] $221 \quad \overset{12231513221}{(12)} \quad 12231^{1}$

$T=20,\ \Delta=-48,\ \det=1,\ (n_+,n_-,n_0)=(1,20,0)$

[5] $1 \quad \underset{1^{1}2231513221}{\overset{21}{(12)}} \quad 12231^{1}$

$T=20,\ \Delta=-48,\ \det=4,\ (n_+,n_-,n_0)=(1,20,0)$

[6] $221 \quad \overset{1^{1}2231513221}{(12)} \quad 12231^{1}$

$T=20,\ \Delta=-48,\ \det=4,\ (n_+,n_-,n_0)=(1,20,0)$

$T=22$, Ext $=\mathfrak{e}_6$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ (6)

[1] $1 \quad \underset{1^{1}231513221}{\overset{1}{(12)}} \quad 122315131$

$T=22,\ \Delta=-100,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[2] $1 \quad \underset{1231513221}{\overset{1}{(12)}} \quad 122315131^{1}$

$T=22,\ \Delta=-100,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[3] $21 \quad \overset{1^{1}231513221}{(12)} \quad 122315131$

$T=22,\ \Delta=-100,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[4] $21 \quad \overset{1231513221}{(12)} \quad 122315131^{1}$

$T=22,\ \Delta=-100,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[5] $1 \quad \underset{1^{1}231513221}{\overset{1}{(12)}} \quad 122315131^{1}$

$T=22,\ \Delta=-100,\ \det=4,\ (n_+,n_-,n_0)=(1,22,0)$

[6] $21 \quad \overset{1^{1}231513221}{(12)} \quad 122315131^{1}$

$T=22,\ \Delta=-100,\ \det=4,\ (n_+,n_-,n_0)=(1,22,0)$

$T=22$, Ext $=\mathfrak{f}_4$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ (6)

[1] $1 \quad \underset{1^{1}231513221}{\overset{1}{(12)}} \quad 122315131$

$T=22,\ \Delta=-74,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[2] $1 \quad \underset{1231513221}{\overset{1}{(12)}} \quad 122315131^{1}$

$T=22,\ \Delta=-74,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[3] $21 \quad \overset{1231513221}{(12)} \quad 122315131^{1}$

$T=22,\ \Delta=-74,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[4] $21 \quad \overset{1^{1}231513221}{(12)} \quad 122315131$

$T=22,\ \Delta=-74,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[5] $1 \quad \underset{1^{1}231513221}{\overset{1}{(12)}} \quad 122315131^{1}$

$T=22,\ \Delta=-74,\ \det=4,\ (n_+,n_-,n_0)=(1,22,0)$

[6] $21 \quad \overset{1^{1}231513221}{(12)} \quad 122315131^{1}$

$T=22,\ \Delta=-74,\ \det=4,\ (n_+,n_-,n_0)=(1,22,0)$

$T=22$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus\mathfrak{su}_3$ (6)

[1] $1 \quad \underset{1231513221}{\overset{1}{(12)}} \quad 122315131^{1}$

$T=22,\ \Delta=-50,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[2] $1\ \underset{1^{1}231513221}{\overset{1}{(12)}}\ 122315131$

$T=22,\ \Delta=-50,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[3] $21\ \overset{1^{1}231513221}{(12)}\ 122315131$

$T=22,\ \Delta=-50,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[4] $21\ \overset{1231513221}{(12)}\ 122315131^{1}$

$T=22,\ \Delta=-50,\ \det=1,\ (n_+,n_-,n_0)=(1,22,0)$

[5] $1\ \underset{1^{1}231513221}{\overset{1}{(12)}}\ 122315131^{1}$

$T=22,\ \Delta=-50,\ \det=4,\ (n_+,n_-,n_0)=(1,22,0)$

[6] $21\ \overset{1^{1}231513221}{(12)}\ 122315131^{1}$

$T=22,\ \Delta=-50,\ \det=4,\ (n_+,n_-,n_0)=(1,22,0)$

$T=23$, Ext $=\mathfrak{e}_6$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_3$ (6)

[1] $1\ \underset{12231513221}{\overset{1}{(12)}}\ 122315131^{1}$

$T=23,\ \Delta=-71,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[2] $1\ \underset{1^{1}2231513221}{\overset{1}{(12)}}\ 122315131$

$T=23,\ \Delta=-71,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[3] $21\ \overset{12231513221}{(12)}\ 122315131^{1}$

$T=23,\ \Delta=-71,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[4] $21\ \overset{1^{1}2231513221}{(12)}\ 122315131$

$T=23,\ \Delta=-71,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[5] $1\ \underset{1^{1}2231513221}{\overset{1}{(12)}}\ 122315131^{1}$

$T=23,\ \Delta=-71,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

[6] $21\ \overset{1^{1}2231513221}{(12)}\ 122315131^{1}$

$T=23,\ \Delta=-71,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

$T=23$, Ext $=\mathfrak{f}_4$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_3$ (6)

[1] $1\ \underset{12231513221}{\overset{1}{(12)}}\ 122315131^{1}$

$T=23,\ \Delta=-45,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[2] $1\ \underset{1^{1}2231513221}{\overset{1}{(12)}}\ 122315131$

$T=23,\ \Delta=-45,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[3] $21\ \overset{1^{1}2231513221}{(12)}\ 122315131$

$T=23,\ \Delta=-45,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[4] $21\ \overset{12231513221}{(12)}\ 122315131^{1}$

$T=23,\ \Delta=-45,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[5] $1\ \underset{1^{1}2231513221}{\overset{1}{(12)}}\ 122315131^{1}$

$T=23,\ \Delta=-45,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

[6] $21\ \overset{1^{1}2231513221}{(12)}\ 122315131^{1}$

$T=23,\ \Delta=-45,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

$T=23$, **Ext** $=\mathfrak{f}_4$, **LST** $=\mathfrak{e}_7^2\oplus\mathfrak{e}_7'\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus(\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2)^3$ **(6)**

[1] $1\ \overset{1321}{8}\ 12321812321712321^{1}$

$T=23,\ \Delta=-71,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[2] $1\ \overset{1^{1}321}{8}\ 12321812321712321$

$T=23,\ \Delta=-71,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[3] $1\ \overset{12321}{8}\ 1232181232171231^{1}$

$T=23,\ \Delta=-71,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[4] $1\ \overset{1^{1}2321}{8}\ 1232181232171231$

$T=23,\ \Delta=-71,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[5] $1\ \overset{1^{1}321}{8}\ 12321812321712321^{1}$

$T=23,\ \Delta=-71,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

[6] $1\ \overset{1^{1}2321}{8}\ 1232181232171231^{1}$

$T=23,\ \Delta=-71,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

$T=23$, **Ext** $=\mathfrak{so}_8$, **LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_3$ **(6)**

[1] $1\ \underset{12231513221}{\overset{1}{(12)}}\ 122315131^{1}$

$T=23,\ \Delta=-21,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[2] $1\ \underset{1^{1}2231513221}{\overset{1}{(12)}}\ 122315131$

$T=23,\ \Delta=-21,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[3] $21\ \overset{12231513221}{(12)}\ 122315131^{1}$

$T=23,\ \Delta=-21,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[4] $21\ \overset{1^{1}2231513221}{(12)}\ 122315131$

$T=23,\ \Delta=-21,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[5] $1\ \underset{1^{1}2231513221}{\overset{1}{(12)}}\ 122315131^{1}$

$T=23,\ \Delta=-21,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

[6] $21\ \overset{1^{1}2231513221}{(12)}\ 122315131^{1}$

$T=23,\ \Delta=-21,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

$T=23$, **Ext** $=\mathfrak{so}_8$, **LST** $=\mathfrak{e}_7^2\oplus\mathfrak{e}_7'\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus(\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2)^3$ **(6)**

[1] $1\ \overset{1321}{8}\ 12321812321712321^{1}$

$T=23,\ \Delta=-47,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[2] $1\ \overset{1^{1}321}{8}\ 12321812321712321$

$T=23,\ \Delta=-47,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[3] $1\ \overset{1^{1}2321}{8}\ 1232181232171231$

$T=23,\ \Delta=-47,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[4] $1\ \overset{12321}{8}\ 1232181232171231^{1}$

$T=23,\ \Delta=-47,\ \det=-1,\ (n_+,n_-,n_0)=(1,23,0)$

[5] $1\ \overset{1^{1}321}{8}\ 12321812321712321^{1}$

$T=23,\ \Delta=-47,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

[6] $1\ \overset{1^{1}2321}{8}\ 1232181232171231^{1}$

$T=23,\ \Delta=-47,\ \det=-4,\ (n_+,n_-,n_0)=(1,23,0)$

$T=24$, **Ext** $=\mathfrak{e}_6$, **LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ **(6)**

[1] $1\overset{1}{\underset{12231513221}{(12)}}1223151321^{1}$

$T=24,\ \Delta=-43,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[2] $1\overset{1}{\underset{1^{1}2231513221}{(12)}}1223151321$

$T=24,\ \Delta=-43,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[3] $21\overset{1^{1}2231513221}{(12)}1223151321$

$T=24,\ \Delta=-43,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[4] $21\overset{12231513221}{(12)}1223151321^{1}$

$T=24,\ \Delta=-43,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[5] $1\overset{1}{\underset{1^{1}2231513221}{(12)}}1223151321^{1}$

$T=24,\ \Delta=-43,\ \det=4,\ (n_+,n_-,n_0)=(1,24,0)$

[6] $21\overset{1^{1}2231513221}{(12)}1223151321^{1}$

$T=24,\ \Delta=-43,\ \det=4,\ (n_+,n_-,n_0)=(1,24,0)$

$T=24$, **Ext** $=\mathfrak{e}_7$, **LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ **(6)**

[1] $1\overset{1}{\underset{12231513221}{(12)}}1223151321^{1}$

$T=24,\ \Delta=-98,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[2] $1\overset{1}{\underset{1^{1}2231513221}{(12)}}1223151321$

$T=24,\ \Delta=-98,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[3] $21\overset{12231513221}{(12)}1223151321^{1}$

$T=24,\ \Delta=-98,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[4] $21\overset{1^{1}2231513221}{(12)}1223151321$

$T=24,\ \Delta=-98,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[5] $1\overset{1}{\underset{1^{1}2231513221}{(12)}}1223151321^{1}$

$T=24,\ \Delta=-98,\ \det=4,\ (n_+,n_-,n_0)=(1,24,0)$

[6] $21\overset{1^{1}2231513221}{(12)}1223151321^{1}$

$T=24,\ \Delta=-98,\ \det=4,\ (n_+,n_-,n_0)=(1,24,0)$

$T=24$, **Ext** $=\mathfrak{e}_7'$, **LST** $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ **(6)**

[1] $1\overset{1}{\underset{1^{1}2231513221}{(12)}}1223151321$

$T=24,\ \Delta=-70,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[2] $1\overset{1}{\underset{12231513221}{(12)}}1223151321^{1}$

$T=24,\ \Delta=-70,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[3] $21\overset{1^{1}2231513221}{(12)}1223151321$

$T=24,\ \Delta=-70,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[4] $21\overset{12231513221}{(12)}1223151321^{1}$

$T=24,\ \Delta=-70,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[5] $1\overset{1}{\underset{1^{1}2231513221}{(12)}}1223151321^{1}$

$T=24,\ \Delta=-70,\ \det=4,\ (n_+,n_-,n_0)=(1,24,0)$

[6] $21\overset{1^{1}2231513221}{(12)}1223151321^{1}$

$T=24,\ \Delta=-70,\ \det=4,\ (n_+,n_-,n_0)=(1,24,0)$

$T=24$, Ext $=\mathfrak{f}_4$, LST $=\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2$ (6)

[1] $1\underset{1^{1}2231513221}{\overset{1}{(12)}}1223151321$

$T=24,\ \Delta=-17,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[2] $1\underset{12231513221}{\overset{1}{(12)}}1223151321^{1}$

$T=24,\ \Delta=-17,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[3] $21\overset{12231513221}{(12)}1223151321^{1}$

$T=24,\ \Delta=-17,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[4] $21\overset{1^{1}2231513221}{(12)}1223151321$

$T=24,\ \Delta=-17,\ \det=1,\ (n_+,n_-,n_0)=(1,24,0)$

[5] $1\underset{1^{1}2231513221}{\overset{1}{(12)}}1223151321^{1}$

$T=24,\ \Delta=-17,\ \det=4,\ (n_+,n_-,n_0)=(1,24,0)$

[6] $21\overset{1^{1}2231513221}{(12)}1223151321^{1}$

$T=24,\ \Delta=-17,\ \det=4,\ (n_+,n_-,n_0)=(1,24,0)$

$T=26$, Ext $=\mathfrak{e}_6$, LST $=\mathfrak{e}_6^2\oplus\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_3^3$ (6)

[1] $\overset{1}{1}61316131513221(12)122315131^{1}$

$T=26,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[2] $\overset{1}{1}6131513221(12)1223151316131^{1}$

$T=26,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[3] $21513221(12)12231513161316131^{1}$

$T=26,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[4] $1\overset{1^{1}3161316131513221(12)12231}{\qquad 5 \qquad}1$

$T=26,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[5] $2161316131513221(12)122315131^{1}$

$T=26,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[6] $216131513221(12)1223151316131^{1}$

$T=26,\ \Delta=-147,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

$T=26$, Ext $=\mathfrak{f}_4$, LST $=\mathfrak{e}_6^2\oplus\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_3^3$ (6)

[1] $\overset{1}{1}61316131513221(12)122315131^{1}$

$T=26,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[2] $\overset{1}{1}6131513221(12)1223151316131^{1}$

$T=26,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[3] $21513221(12)12231513161316131^{1}$

$T=26,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[4] $1\overset{1^{1}3161316131513221(12)12231}{\qquad 5 \qquad}1$

$T=26,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[5] $2161316131513221(12)122315131^{1}$

$T=26,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[6] $216131513221(12)1223151316131^{1}$

$T=26,\ \Delta=-121,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

$T=26$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{e}_6^2\oplus\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus\mathfrak{su}_3^3$ (6)

[1] $1\overset{1}{6}1316131513221(12)122315131^{1}$

$T=26,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[2] $1\overset{1}{6}131513221(12)1223151316131^{1}$

$T=26,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[3] $21513221(12)12231513161316131^{1}$

$T=26,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[4] $1\,\overset{1^{1}3161316131513221(12)12231}{5}\,1$

$T=26,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[5] $2161316131513221(12)122315131^{1}$

$T=26,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

[6] $216131513221(12)1223151316131^{1}$

$T=26,\ \Delta=-97,\ \det=1,\ (n_+,n_-,n_0)=(1,26,0)$

$T=29$, Ext $=\mathfrak{f}_4$, LST $=\mathfrak{e}_7^3\oplus\mathfrak{e}_7'\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus(\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2)^4$ (6)

[1] $1\overset{1321}{8}12321812321812321712321^{1}$

$T=29,\ \Delta=-41,\ \det=-1,\ (n_+,n_-,n_0)=(1,29,0)$

[2] $1\overset{1^{1}321}{8}12321812321812321712321$

$T=29,\ \Delta=-41,\ \det=-1,\ (n_+,n_-,n_0)=(1,29,0)$

[3] $1\overset{12321}{8}1232181232181232171231^{1}$

$T=29,\ \Delta=-41,\ \det=-1,\ (n_+,n_-,n_0)=(1,29,0)$

[4] $1\overset{1^{1}2321}{8}1232181232181232171231$

$T=29,\ \Delta=-41,\ \det=-1,\ (n_+,n_-,n_0)=(1,29,0)$

[5] $1\overset{1^{1}321}{8}12321812321812321712321^{1}$

$T=29,\ \Delta=-41,\ \det=-4,\ (n_+,n_-,n_0)=(1,29,0)$

[6] $1\overset{1^{1}2321}{8}1232181232181232171231^{1}$

$T=29,\ \Delta=-41,\ \det=-4,\ (n_+,n_-,n_0)=(1,29,0)$

$T=29$, Ext $=\mathfrak{so}_8$, LST $=\mathfrak{e}_7^3\oplus\mathfrak{e}_7'\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus(\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2)^4$ (6)

[1] $1\overset{1321}{8}12321812321812321712321^{1}$

$T=29,\ \Delta=-17,\ \det=-1,\ (n_+,n_-,n_0)=(1,29,0)$

[2] $1\overset{1^{1}321}{8}12321812321812321712321$

$T=29,\ \Delta=-17,\ \det=-1,\ (n_+,n_-,n_0)=(1,29,0)$

[3] $1\overset{12321}{8}1232181232181232171231^{1}$

$T=29,\ \Delta=-17,\ \det=-1,\ (n_+,n_-,n_0)=(1,29,0)$

[4] $1\overset{1^{1}2321}{8}1232181232181232171231$

$T=29,\ \Delta=-17,\ \det=-1,\ (n_+,n_-,n_0)=(1,29,0)$

[5] $1\overset{1^{1}321}{8}12321812321812321712321^{1}$

$T=29,\ \Delta=-17,\ \det=-4,\ (n_+,n_-,n_0)=(1,29,0)$

[6] $1\overset{1^{1}2321}{8}1232181232181232171231^{1}$

$T=29,\ \Delta=-17,\ \det=-4,\ (n_+,n_-,n_0)=(1,29,0)$

$T=32$, **Ext** $=\mathfrak{f}_4$, **LST** $=\mathfrak{e}_7^2\oplus\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus(\mathfrak{su}_2\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2$ **(6)**

[1] $132181232181231513221(12)1223151321^{1}$
$T=32,\ \Delta=-71,\ \det=1,\ (n_+,n_-,n_0)=(1,32,0)$
[2] $1231513221(12)122315132181232181231^{1}$
$T=32,\ \Delta=-71,\ \det=1,\ (n_+,n_-,n_0)=(1,32,0)$
[3] $1232181231513221(12)122315132181231^{1}$
$T=32,\ \Delta=-71,\ \det=1,\ (n_+,n_-,n_0)=(1,32,0)$
[4] $132181231513221(12)1223151321812321^{1}$
$T=32,\ \Delta=-71,\ \det=1,\ (n_+,n_-,n_0)=(1,32,0)$
[5] $1^{1}32181232181231513221(12)1223151321^{1}$
$T=32,\ \Delta=-71,\ \det=4,\ (n_+,n_-,n_0)=(1,32,0)$
[6] $1^{1}32181231513221(12)1223151321812321^{1}$
$T=32,\ \Delta=-71,\ \det=4,\ (n_+,n_-,n_0)=(1,32,0)$

$T=32$, **Ext** $=\mathfrak{so}_8$, **LST** $=\mathfrak{e}_7^2\oplus\mathfrak{e}_8\oplus\mathfrak{f}_4^2\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^2\oplus(\mathfrak{su}_2\oplus\mathfrak{g}_2)^3\oplus\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2$ **(6)**

[1] $1231513221(12)122315132181232181231^{1}$
$T=32,\ \Delta=-47,\ \det=1,\ (n_+,n_-,n_0)=(1,32,0)$
[2] $132181232181231513221(12)1223151321^{1}$
$T=32,\ \Delta=-47,\ \det=1,\ (n_+,n_-,n_0)=(1,32,0)$
[3] $132181231513221(12)1223151321812321^{1}$
$T=32,\ \Delta=-47,\ \det=1,\ (n_+,n_-,n_0)=(1,32,0)$
[4] $1232181231513221(12)122315132181231^{1}$
$T=32,\ \Delta=-47,\ \det=1,\ (n_+,n_-,n_0)=(1,32,0)$
[5] $1^{1}32181232181231513221(12)1223151321^{1}$
$T=32,\ \Delta=-47,\ \det=4,\ (n_+,n_-,n_0)=(1,32,0)$
[6] $1^{1}32181231513221(12)1223151321812321^{1}$
$T=32,\ \Delta=-47,\ \det=4,\ (n_+,n_-,n_0)=(1,32,0)$

$T=35$, **Ext** $=\mathfrak{f}_4$, **LST** $=\mathfrak{e}_7^4\oplus\mathfrak{e}_7'\oplus\mathfrak{su}_2\oplus\mathfrak{g}_2\oplus(\mathfrak{su}_2\oplus\mathfrak{so}_7\oplus\mathfrak{su}_2)^5$ **(6)**

[1] $1\overset{1^{1}321}{8}1232181232181232181232171232 1$
$T=35,\ \Delta=-11,\ \det=-1,\ (n_+,n_-,n_0)=(1,35,0)$
[2] $1\overset{1321}{8}12321812321812321812321712321^{1}$
$T=35,\ \Delta=-11,\ \det=-1,\ (n_+,n_-,n_0)=(1,35,0)$
[3] $1\overset{1^{1}2321}{8}1232181232181232181232171231$
$T=35,\ \Delta=-11,\ \det=-1,\ (n_+,n_-,n_0)=(1,35,0)$
[4] $1\overset{12321}{8}1232181232181232181232171231^{1}$
$T=35,\ \Delta=-11,\ \det=-1,\ (n_+,n_-,n_0)=(1,35,0)$
[5] $1\overset{1^{1}321}{8}12321812321812321812321712321^{1}$
$T=35,\ \Delta=-11,\ \det=-4,\ (n_+,n_-,n_0)=(1,35,0)$
[6] $1\overset{1^{1}2321}{8}1232181232181232181232171231^{1}$
$T=35,\ \Delta=-11,\ \det=-4,\ (n_+,n_-,n_0)=(1,35,0)$

$T=48$, **Ext** $=\mathfrak{f}_4$, **LST** $=\mathfrak{e}_8^4\oplus\mathfrak{f}_4^3\oplus(\mathfrak{sp}_1\oplus\mathfrak{g}_2)^7$ **(6)**

[1] $1\overset{221}{(12)}12231513221\overset{1}{(12)}12231513221(12)12231513221\overset{1}{(12)}12231^{1}$
$T=48,\ \Delta=-144,\ \det=1,\ (n_+,n_-,n_0)=(1,48,0)$
[2] $21\overset{21}{(12)}12231513221\overset{1}{(12)}12231513221(12)12231513221\overset{1}{(12)}12231^{1}$

$T = 48,\ \Delta = -144,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[3] $1\overset{2221}{(12)}12231513221(12)12231513221\overset{1}{(12)}12231513221(12)12231^{1}$

$T = 48,\ \Delta = -144,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[4] $1\ \overset{1^{1}3221}{(12)}\ 12231513221(12)12231513221\overset{1}{(12)}12231513221(12)1222$

$T = 48,\ \Delta = -144,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[5] $21\overset{221}{(12)}12231513221(12)12231513221\overset{1}{(12)}12231513221(12)12231^{1}$

$T = 48,\ \Delta = -144,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[6] $22221(12)12231513221(12)12231513221\overset{1}{(12)}12231513221(12)12231^{1}$

$T = 48,\ \Delta = -144,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

$T = 48$**, Ext** $= \mathfrak{so}_8$**, LST** $= \mathfrak{e}_8^4 \oplus \mathfrak{f}_4^3 \oplus (\mathfrak{sp}_1 \oplus \mathfrak{g}_2)^7$ **(6)**

[1] $1\overset{221}{(12)}12231513221\overset{1}{(12)}12231513221(12)12231513221\overset{1}{(12)}12231^{1}$

$T = 48,\ \Delta = -120,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[2] $21\overset{21}{(12)}12231513221\overset{1}{(12)}12231513221(12)12231513221\overset{1}{(12)}12231^{1}$

$T = 48,\ \Delta = -120,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[3] $1\overset{2221}{(12)}12231513221(12)12231513221\overset{1}{(12)}12231513221(12)12231^{1}$

$T = 48,\ \Delta = -120,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[4] $1\ \overset{1^{1}3221}{(12)}\ 12231513221(12)12231513221\overset{1}{(12)}12231513221(12)1222$

$T = 48,\ \Delta = -120,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[5] $21\overset{221}{(12)}12231513221(12)12231513221\overset{1}{(12)}12231513221(12)12231^{1}$

$T = 48,\ \Delta = -120,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$

[6] $22221(12)12231513221(12)12231513221\overset{1}{(12)}12231513221(12)12231^{1}$

$T = 48,\ \Delta = -120,\ \det = 1,\ (n_+, n_-, n_0) = (1, 48, 0)$